

TRANSPORTATION ENGINEERING LAB
(BCE23411P)
LABORATORY MANUAL



DEPARTMENT OF CIVIL ENGINEERING

**GIRIJANANDA CHOWDHURY INSTITUTE OF
MANAGEMENT AND TECHNOLOGY**

AZARA , ASSAM – 781017

SYLLABUS

Course Code	Course Title	Hours per week L- T - P	Credit
CE181514	Transportation Engineering Lab	0 -0 - 2	1

1. Sieve analysis [as per IS 2386 (Part I)-1963] and blending of aggregates by Triangulation and Rothfuchs Method
2. Combined Flakiness and Elongation Index test [as per IS 2386 (Part I)-1963]
3. Specific Gravity test of coarse and fine aggregates [as per IS 2386 -(Part III)1963]
4. Aggregate Impact Value test [as per IS 2386 (Part IV)-1963]
5. Los Angeles Abrasion Value test [as per IS 2386 (Part IV)-1963]
6. Aggregate Crushing value Test [as per IS 2386 (Part IV)-1963]
7. California Bearing Ratio (CBR) test [as per IS 2386 (Part XVI)-1987]
8. Dynamic Cone Penetrometer test [as per ASTM D6951/D6951M-09]
9. Penetration test on bitumen [as per IS: 1203-1978]
10. Ductility test on bitumen [as per IS: 1208 –1978]
11. Ring and Ball test for softening point determination [as per IS 1205-1978]
12. Flash and fire point determination using Pensky-Martens apparatus [as per IS: 1209 1978]
13. Striping value test of aggregates [as per IS: 6241-1971]
14. Saybolt Viscosity test on bitumen emulsion [as per ASTM D7496-11]
15. Marshal Stability test on bituminous mixes to determine optimum binder content [as per ASTMD6927, ASTM-D5581]

Experiment no: 1

Title: Sieve analysis Aggregate.

Aim : To determine the Sieve analysis of coarse aggregate.

Apparatus Equipment:

- Sieve type: 80 mm, 63 mm, 40 mm, 20 mm, 16 mm, 12.5 mm and 10 mm. (Square hole perforated plate, 30 cm diameter)
- Fine mesh, wire cloth, 20 cm diameter 10 mm, 4.75 mm, 2.36 mm, 1.18 mm, 600 micron, 300 Micron, 150 micron, 75 micron.
- Balance (to weight 10 kg) brushes with fined camel hair.

(a) Sieve analysis results for fine aggregate

I.S sieve size	Weight of aggregate retained	Total weight				
		sum	Average sum	% retained	Cumulative % retained	Percent passing
1	2	3	4	5	6	7
10mm						
4.75mm						
2.36mm						
1.18mm						
500 mic						
4.25 mic						
150 mic						
75 mic						
pan						

$$\text{Fineness modulus} = \frac{\sum \text{Cumulative \%}}{100}$$

(b) Sieve analysis results for coarse aggregate

I.S sieve size	Weight of aggregate retained			Total weight				
	I	II	III	Sum	Average sum	% retained	Cumulative % retained	Percent passing
1	2	3	4	5	6	7	8	9
2								
3								
4								
5								

6						
7						
8						

Fineness modulus of CA = $\frac{\sum \text{Cumulative \%}}{100}$

FM of CA =

I.S SIEVE DESIGNATION	PERCENTAGE PASSING FOR			
	Grading zoneI	Grading zoneII	Grading zoneIII	Grading zoneIV
10 mm	100	100	100	100
4.75 mm	9-100	9-100	90-100	95-100
2.36 mm	60-95	75-100	85-100	95-100
1.18 mm	30-70	55-90	75-100	90-100
600 μ	15-34	35-59	60-79	80-100
300 μ	5-20	8-30	12-40	15-50
150 μ	0-10	0-10	0-19	0-15

Here,

Fine sand → 2.2 to 2.6

Medium sand → 2.6 to 2.9

Coarse sand → 2.9 to 3.2

Experiment No:2

Title: Flakiness Index Test.

Aim: To determine the Flakiness Index.

Apparatus/Equipment : Balance, Sieve (63, 50,40, 31.5, 25, 20, 16, 12.6,10, 6.13mm), shaker etc.

Theory: Flakiness index of aggregates is the percentage by weight of particles whose least dimension (thickness) is less than three fifth of their mean dimension is the average size of aggregates. The test is applicable to size larger than 6.3 mm.

Flakiness index: $\text{Weight of the material passing through thickness gauge} \times 100$ Total weight of aggregate.

Procedure:

1. Take some aggregate and sieve them properly through the sieve at (63 mm to 6.3 mm). The taken quantity must be sufficient to provide minimum 200 sieved pieces for test.
2. Separate the aggregate by their sizes and find the weight of aggregate of each size(w).
3. Try to pass each aggregate through corresponding slot in the thickness gauge.
4. Weight all the pieces which passes through the particular slot.

Observation:

Sl no.	Passing IS sieve	Retained IS seive	Thickness of the slot	Weight of the material corresponding size (W)	Weight of the material through slot (mm) kg
1	63	50	33.9		
2	50	40	27.0		
3	40	31.5	19.5		
4	31.5	25	16.95		
5	25	20	13.5		
6	20	16	10.5		
7	16	12.5	8.56		
8	12.5	10	6.76		
9	10	6.3	4.89		

Calculation:

a) Total weight of material. $W = W_1 + W_2 + W_3 + \dots$

b) Weight of material passing through thickness gauge, $w = w_1 + w_2 + w_3 + \dots$

c) Flakiness index = $w/W \times 100$

Title: Elongation index test.

Aim: To determine the Elongation index.

Apparatus/Equipment: Balance, sieve (63, 50, 40, 31.5, 25, 20, 16, 12.6, 10, 6.13 mm), shaker, etc.

Theory:

Elongation index of aggregates is the percentage by weight of particles whose greatest dimension is greater than one and four fifth (1.8) times their mean dimension. This test is application to larger than 6.3 mm.

$$\text{Elongation index} = \frac{\text{Weight of material retained through thickness gauge}}{\text{Total weight of aggregate}} \times 100$$

Procedure:

1. Sieve the aggregates through the sieve set and determine the weight of aggregates of each size separately (W).
2. Try to pass each piece through corresponding gauge size.
3. Weight all the material retained by length gauge.

Observation:

(A)

Sl. no.	Passing IS sieve.	Retained on IS sieve.	Thickness of slot (mm).	Weight of material of corresponding size (W).	Weight of material retained (W).
1	63	50	33.9		
2	50	40	27.0		
3	40	31.5	19.5		
4	31.5	25	61.95		
5	25	20	13.5		
6	20	16	10.8		
7	16	12.5	8.56		
8	12.5	10	6.75		
9	10	6.3	4.89		

(B)

Sl. no	Passing through IS sieve.	Retained on IS sieve.	Length of gauge.	Weight of material.	Weight of material by length of gauge.
1	63	50	-	-	-
2	50	40	-	-	-
3	40	31.5	81.0	-	-
4	31.5	25	-	-	-
5	25	20	40.5	-	-
6	20	16	32.4	-	-
7	16	12.5	25.6	-	-
8	12.5	10	20.2	-	-
9	10	6.3	14.7	-	-

Calculation:(A) Total weight of material $W_1+W_2+W_3+\dots\dots\dots =$ (B) Weight of material retained by length gauge $w =$ (C) Elongation index $= \frac{w}{W}$ **Results:**Elongation index of aggregates $=$

Experiment no:3

Title: Specific Gravity and Water Absorption of Aggregate

Aim : To determine Specific Gravity and Water Absorption of Aggregate.

Apparatus/Equipment:

1. Balance: A balance or scale of capacity not less than 3kg readable and accurate to 0.5g and of such a type as to permit the weighting of the vessel containing aggregate and water.
2. Oven: Well ventilated and fitted with control to maintain a temperature of 100°C to 110°C.
3. Wide mouthed glass vessel such as jar of about 1.5 liter capacity, with a flat ground lip and a plane ground disc of plate glass to cover it giving a virtually water tight fit.
4. Clod: Dry soft absorbent clothes, each not less than 72cm x 45 cm.
5. Tray: Sallow tray of area not less than 325 cm².
6. Container: An air tight container, large enough to take the sample.
7. Wine basket of not more than 6.3 mm mesh or a perforated container of convenient size, preferably chromium plated and polished with wire hangers not thicker than 1mm for suspending it from the balance.
8. Pycnometer.

Theory: Specific gravity is the ratio of the weight of aggregate maintained for $24 \pm \frac{1}{2}$ hours at temperature 100°C to 110°C to the weight of equal volume of water displaced by saturated surface dry aggregate. In other word it means volume of water displaced is equal to the volume of solid matter of the aggregate and volume of all pores both impermeable and permeable.

Apparent specific gravity is the ratio of the weight of aggregate dried and maintained for 24 hours $\pm \frac{1}{2}$ hours at temperature 100°C to 110°C to the weight of equal volume of water displaced by the dry aggregate. The volume of water displaced is therefore, equal to the water displaced by solid matter and only impermeable pores.

It is the specific gravity calculated on the saturated surface dry basis that is normally used in the calculation of absolute volumes of concrete and yield of concrete.

Majority of aggregates have the values between 2.6 to 2.9. Values lower than 2.6 however, do not necessarily mean that the aggregate is of lower quantity. No minimum limit has been specified for specific gravity. In certain constructions like gravity dams where the weight of the concrete is important from the point of view of stability, minimum limit is specified.

Specific gravity is useful for calculating void content in aggregate. If the aggregate is dry, it absorbs water which doesn't take part in the reactions and also in lubrication of particles. Thus workability is likely to be reduced and w/c ratio is also altered. It is therefore, Always necessary to make allowance for water absorption while calculating total water to be added to the mixes.

The absorption value is the difference in weight between the saturated surface dry aggregate

and very dry sample expressed as percentage of dry weight of aggregate.

Specific gravity of aggregate can be used the same way as bulk density for judging quality by comparison with normal specific gravity for that type of aggregate.

Procedure:

1. Take two kg of sample.
2. The sample is screened on a 10 mm IS sieve, thoroughly washed to remove fine particles of dust, danted, then placed in the wire basket immersed in distilled water and allowed to remain immersed at a temperature of 22°C to 32°C for 24±½ hours.
3. Soon after immersion and again at the end of the soaking period air entrapped in or bubbles on the surface of the aggregate should be removed by gentle agitation. This is achieved by rapid clock-wise and anti-clockwise rotation of vessel between the operator's hands.
4. The vessel is over filled by adding distilled water and the plane ground glass disc slid over the Mouth so as to ensure that no air is trapped in the vessel. The shall be dried on the outside and weighed (A).
5. The vessel is emptied and the aggregate is allowed to drain out. Refill the vessel with distilled water.
6. Slide the glass disc in position as before. The vessel is dried and weighed (B).
7. The aggregate is placed on a dry cloth and gently surface dried with the cloth transferring it to a second dry cloth when the first will remove no further moisture.
8. It is the spread out not more than one stone deep on the second cloth and left exposed to atmosphere away from direct sunlight or any other source of heat for no less than 10 minutes or until it appears to be completely surface dry (which with some aggregates may take an hour or more) the aggregate is turned over at least once during this period and a gentle current of unheated air may be used after first 10 minutes to accelerates the drying of the difficult aggregate.
9. The aggregate shall be weighed (C).
10. The aggregate is placed in the oven in the shallow tray, at a temperature of 100° C to 110°C for 24 ±½ hours. It is then cooled in air tight container and weighed (D).
11. Determine specific gravity and water absorption value.

Calculation:

- 1.Weight of saturated and surface dry aggregate (C)g
2. Weight of pycnometer sample water (A)g
- 3.Weight of pycnometer water (B)g
4. Weight of men dry sample (D)g
5. Specific gravity $= \frac{C-D}{D} \times 100\%$

.....

Experiment no:4

Title : Impact Test value of Coarse aggregate

Aim: To determine the toughness of road stone materials by impact test.

Introduction:

Toughness is the property of a material to resist impact. Due to traffic loads, the road stones are subjected to the pounding action or impact and there is possibility of stones breaking into smaller Pieces. The road stones should therefore be tough enough to resist fracture under impact. A test designed to evaluate the toughness of stones i.e., the resistance of the fracture under repeated impacts may be called an impact test for road stones.

Apparatus:

- (a) Impact testing machine.
- (b) Measure: A cylindrical metal measure having internal diameter 75mm and depth 50mm for measuring aggregates.
- (c) Tamping rod: A straight metal tamping rod of circular cross section, 10mm in diameter and 230mm long rounded at one end.
- (d) Sieve IS sieve of sizes 12.5mm, 10mm, and 2.36mm for sieving the aggregates.
- (e) Balance: A balance of capacity not less than 500 gm to weight accurate up to 0.1 gm.
- (f) Oven: A thermostatically controlled drying oven capable of maintaining constant temperature between 100°C to 110°C.

Procedure:

The test sample consists of aggregates passing 12.5mm sieve and retained on 10mm sieve and dried in an oven for four hours at a temperature 100°C to 110°C and cooled. Test aggregates are filled up to about one-third full in the cylindrical measure and tamped 25 times with rounded end of the tamping rod. Further quantity of aggregates is then added up to two-third full in the cylinder and 25 stocks of the tamping rod are given. The measure is now filled with the aggregate to over flow temped 25 times. The surplus aggregates are struck off using the tamping rod as straight edge. The net weight of the aggregates in the measure is determined to the nearest gram and this weight of the aggregates is used for carrying out duplicate test on the same material. The impact machine is placed with its bottom plate flat on the floor so that the hammer guide columns are vertical. The cup is fixed firmly in position on the base of the machine and the whole of the test sample from the cylindrical measure is transferred to the cup and compacted by tamping with 25 strokes.

The hammer is raised until its lower face is 380mm above the upper surface of the aggregates in the cop, and allowed to fall freely on the aggregates. The test sample is subjected to a total 15 such blows each being delivered at an interval of not less than one second. The crushed aggregate is then removed from the cup and the whole of it sieved on the 2.36mm sieve until no further significant amount passes. The fraction retained passing the sieve is weighed accurate to 0.1 gm. The fraction retained on the sieve is also weighted and if the total weight of the fractions passing and retained on the sieve is added it should not be less the original weight of the specimen by more than one gram, if the total weight is less than the original by over one gram the results should be discarded and a fresh test made.

Calculations:

The aggregate impact value is expressed as the percentage of the fines formed in terms of the total weight of the sample.

$$\text{Aggregates Impact Value} = \frac{W_2}{W_1} \times 100 \quad \text{Where,}$$

W1= Original weight of the sample.

W2= Weight of fraction passing 2.36 mm IS sieve.

Results: The mean of the three results is reported as the AIV (Aggregate Impact Value) of the specimen to the nearest whole number.

Limits: <10% Exceptionally strong, 10-20% Strong, 20-30% Satisfactory for road surfacing & >35% Weak for road surfacing.

Experiment no : 5

Title : Abrasion test of Coarse aggregate

Aim : To determine the abrasion of coarse aggregate by the use of LOS Angeles machine.

Apparatus : The apparatus shall consist of following: -

LOS Angeles Machine:- The LA abrasion testing machine conforming in all its essential characteristics to the design. The machine shall consist of a hollow steel cylinder, closed at both ends having an inside diameter of 700 mm and an inside length of 500 mm. A removable steel shelf, projecting radially 80 mm into the cylinder and extending its full length shall be mounted along the end element of the interior surface of the cylinder. The shelf should be such thickness and diameter mounted at both bolts. The portion of the shelf shall be such that the distance from the shelf to the opening measured along the circumference of the cylinder in the direction of rotation, shall not be less than 1250 mm.

The abrasion change depending upon the grading of the test samples as described as follows –

Table - I

<i>Grading</i>	<i>No of sphere</i>	<i>Weight of charge</i>
A	12	5000 +25
B	11	4584+25
C	8	3330+20
D	6	2500+15
E	12	5000+25
F	12	5000+25
G	12	5000+25

Table: II – Grading of test sample

Sieve mm passing	Sieve (square holes) retained	Weight in gm of test sample for Grade						
		A	B	C	D	E	F	G
80	63					2500		
63	50					2500		
50	40					2500		
40	25	1250					5000	
25	20	1250					5000	
20	12.5	1250	2500					5000
12.5	10	1250	2500					
10	6.3			2500				
6.3	4.75			2500				
	2.36				500			

Procedure:

The first sample and abrasive charges shall be placed in the LOS Angeles abrasion testing machine and the machine rotated at a speed of 20 to 33 rev/min. For Grading A, B, C, D the machine shall be rotated for 500 revolutions, for grading E, F and G, it shall be rotated for 1000 revolution. The machine shall be rotated in such a direction that the change on the outside surface of the angle. At the completion test, the material shall be discharged from the machine and a preliminary separation of the sample made on a sieve coarser than the 1.70 mm IS sieve. The finer portion shall then be sieved on a 1.70 mm IS sieve in the manners.

Observation:

Original weight of aggregate = W₁

Weight of aggregate sample retained on 1.7 mm sieve = W₂

Weight of aggregate passing 1.7 mm sieve W = W₁ – W₂

Abrasion value = $\frac{W}{W_1} \times 100$

Conclusion:

Los Angeles test is commonly used to evaluate the hardness of aggregate depending upon the Los Angeles abrasion value, the suitability of aggregate for different construction as per IRC specification is given in the table below –

Type of pavement	Maximum permissible value
Water bond macadam sub base coarse	60 %
WBM base coarse with bituminous surfacing	50%
Bituminous bond macadam	50%
WBM surface coarse	40%
Bituminous penetration macadam	40%
Bituminous surface dressing cement concrete surface coarse	35%
Bituminous concrete surface coarse	30%

Experiment no:6

Title: Aggregate crushing value test.

Aim: To determine the aggregate crushing value by compressive testing machine.

Apparatus:

- a) Steel cylinder with open ends, and internal diameter 152mm, square base plate, plunger having a piston of diameter 150mm, with a hole provided across the stem of the plunger so that a rod could be inserted for lifting or placing the plunger in the cylinder.
- b) Cylindrical measure having internal diameter of 115mm and height 180mm.
- c) Steel tamping rod with one rounded end, having a diameter of 16mm and length 450 to 600mm.
- d) Balance of capacity 3 kg with accuracy up to 1gm.
- e) Compressive testing machine capable of applying load of 40 tonnes, at a uniform rate of loading of 4 tonnes per minute.

Procedure:

The aggregate passing 12.5mm IS sieve and retained on 10mm IS sieve is selected for standard test. The aggregate should be in surface dry condition before testing. The aggregate may be dried by heating at a temperature 100°C to 110°C for a period of 4 hours and is tested after being cooled to room temperature.

The cylindrical measure is filled by the test sample of aggregate in three layers of approximately equal depth, each layer being tamped 25 times by the rounded end of the tamping rod. After the third layer is tamped, using the tamping rod as a straight edge levels off the aggregate at the top of the of the cylindrical measure. About 6.5kg of aggregate is required for preparing two test samples. The test sample thus taken is then weighed. The same weight of the sample is taken in the repeat test.

The cylinder of a test apparatus is placed in position on the base plate: one

third of the test sample placed in this cylinder and tamped 25 times by the tamping rod. Similarly, the other two parts of the test specimen are added, each layer being subjected to 25 blows. The total depth of the material in the cylinder after tamping shall however be 100mm. The surface of the aggregates is leveled and the plunger inserted so that it rests on this surface in level position. The cylinder with the test sample and the plunger in position is placed on compression testing machine. Load is then applied through the plunger at a uniform rate of 4 tonnes per minute until the total load is 40 tonnes, and then the load is released. Aggregates including the crushed portion are removed from the cylinder and sieved on a 2.36mm IS sieve. The material, which passes this sieve is collected.

The above crushing test is repeated on second sample of the same weight in accordance with above test procedure. This test is made for the same specimen for taking an average value.

Calculation:

The aggregate crushing value is defined, as a ratio of the weight of fines passing the specified IS sieve to the total weight of the sample expressed as a percentage.

Where,

W1 Total weight of dry sample.

W2 Weight of the portion of crushed material passing 2.36mm IS sieve.

Aggregate crushing value = $W2/W1 \times 100\%$

Results: The mean of the crushing value obtained in the two tests is reported as the aggregate crushing value.

Limits: The aggregate crushing value for cement concrete pavement shall not exceed 30%. The aggregate crushing value for wearing surfaces shall not exceed 45%.

Experiment No 7

Title : California Bearing Ratio (CBR)

Aim: To determine the CBR value of a given soil sample.

Theory: The California Bearing Ratio (CBR) test is a penetration test which was developed by the California State Highway Department, USA for the evaluation of soil-subgrade strength and base course materials strength base course materials strength for flexible pavements.

The CBR is a measure of resistance of a material to penetration of a standard plunger under controlled density and moisture conditions at standard rate. CBR value as defined by IS: 2720 (Part XVI)-1979 is the ratio of the force per unit area required to penetrate a soil mass with a circular plunger of 50 mm diameter at the rate of 1.25 mm/minute, to that required corresponding penetration of a standard material. The CBR test may be controlled in remoulded or undisturbed specimen in the laboratory. The test procedure should be strictly adhered if high degree of reproducibility is desired. Many methods exists today which utilise mainly CBR test values for designing pavements structure. The test is empirical and results cannot be related accurately with any fundamental property of the material. But the test is simple and has been extensively investigated for field correlations of flexible pavement thickness requirement.

Equipment/Apparatus: The apparatus as per IS:2720 (part XVI)-1979 comprises of the following:

1. Loading machine: Any compression machine with a capacity of at least 5000 kg which can operate at constant rate of 1.25mm per minute can be used for this purpose. A plunger of diameter 50mm, proving ring, deflection dial etc.
2. Cylindrical moulds: moulds of 150mm diameter and 175mm height provided with a collar of about 50mm length and a detachable perforated base are used for this purpose. A spacer disk of 148mm and 47.7mm thickness is used to obtain a specimen of exactly 127.3mm height.
3. Spacer disc: A metal disc of 148 mm diameter and 47.7 mm height.
4. Compaction rammer: The material is compacted as specified for the work, either by dynamic compaction or static compaction. The details for dynamic compaction suggested by the ISI are given below:

Type of compaction	Number of layers	Weight of hammer Kg	Free fall cm.	Number of blows
Light compaction	3	2.6	31	56
Heavy compaction	5	4.89	45	56

5. Adjustable stem, perforated plate, tripod and dial gauge: The standard procedure requires that the soil sample before testing should be soaked in water to measure swelling. For these purpose these accessories are required.

6. Surcharge weights: In order to simulate the effect of over lying pavement weight, one annular metal weights each of 2.5 kg and 147mm diameter are placed on top of the specimen, both at the time of soaking and testing the samples, as surcharge.

7. Penetration plunger: A metallic plunger having a diameter of 50 mm and at least 100 mm long.

8. IS sieve of sizes 20mm and 4.75mm, oven, balance, coarse filter etc. equipments a

Procedure:

1. Preparation of test specimen: As per the ISI, the CBR test may be performed on undisturbed soil specimen obtained by fitting a cutting edge to the mould or on remoulded specimens. The material used in the remoulded specimen shall pass a 20-mm IS sieve. About 20 to 45kg of material is sieved and dried through 20mm sieve. If there is noteworthy proportion of material retained on 20mm sieve, allowance for larger materials is made by replacing it by an equal weight of material passing 20mm sieve and 4.75mm sieve. The optimum moisture content and maximum dry density of soil are determined by adopting either IS light compaction (Proctor Compaction) or IS heavy compaction (Modified Proctor Compaction) as per the requirement. In case the most of the sample pass through 4.75mm sieve, then the dry pulverised sample is sieved through 4.75mm sieve and the portion passing this sieve is only used for this test. Remoulded soil specimen may be compacted either by static compaction or by dynamic compaction.

Statically Compacted Specimen: The batch of soil is mixed with water to give the required moisture content. The correct weight of moist soil to obtain the desired density is placed in the mould and compaction is attained by pressing the spacer disc using a compaction or jack. The preparation of a soil specimen by dynamic compaction or ramming is more commonly adopted and is explained below:

Dynamically Compacted Specimen: A representative sample of the soil weighing approximately 5.5kg for granular soil and 4.5 to 5kg weight for fine grained soils is taken and mixed thoroughly with water up to the optimum moisture content or field moisture content if specified so. The spacer disc is placed at the bottom of the mould over the base plate and a coarse filter paper is placed over the spacer discs. The moist soil is to be compacted over this in the mould by compacting either the IS light compaction or the heavy IS compaction.

a) For IS light compaction, the soil to be compacted is divided into three equal parts, the soil is compacted into three equal layers, each of compacted thickness about 44mm by applying 56 evenly distributed blows of the 2.6kg rammer

b) For IS heavy compaction, the soil is divided into five equal parts. The soil is compacted in

five equal layers, each of compacted thickness about 26.5mm by applying 56 evenly

distributed blows of the 4.89kg rammer. After compacting the last layer, the collar is

removed and the excess soil above the top of the mould is evenly trimmed off by means of

the straight edge. It is important to see if the excess soil is to be trimmed off while preparing

each specimen is of thickness about 5mm; If not the weight of soil taken for compacting

each specimen is suitably adjusted for repeat tests so that the thickness of the excess layer

to be trimmed off is about 5mm. Any hole that develops on the surface due to removal of

coarse particles during trimming may be patched up with smaller size materials. Three such

compacted specimens are prepared for the CBR tests. About 100gm of soil samples are

collected from each mould for moisture content determination from trimmed off portions.

2. The clamps are removed and the mould with the compacted soil is lifted leaving below the perforated base plate and spacer disc which is removed. The mould with the compacted soil is weighed. A filter paper is placed on the perforated base plate, the mould with

compacted soil is inverted and placed in position over the base plate (such that the top of the soil sample is now placed over the base plate) and the clamps of the base plate are tightened. Another filter paper is placed on the top surface of the soil sample and the perforated plate with adjustable stem over it. Surcharge weight of 2.5 or 5kg weight are placed over the perforated plate and the whole mould with the weights is placed in a water tank for soaking such that water can enter specimen from both top and bottom. The swell measuring device consisting of the tripod and the dial gauge are placed on the top edge of the mould and the spindle of the dial gauge is placed touching the top of the adjustable low stem of the perforated plate (see fig.2). The initial dial gauge reading is recorded and the test is kept undisturbed in the water tank to allow soaking of the soil specimen for four full days or 96 hours. The final dial gauge reading is noted to measure the expansion or swelling of the specimen due to soaking.

3. The swell measuring assembly is removed, the mould is taken out of the water tank and the sample is allowed to drain in a vertical position for 15 minutes. The surcharge weights, the perforated plate with stem and the filter paper are removed. The mould with the soil sample is removed from the base plate and is weighed again to determine the weight of the water absorbed.

4. The mould with the specimen is clamped over the base plate and the same surcharge weights are placed on the specimen such that the penetration test could be conducted. The mould with the base [plate is placed under the penetration plunger of the loading machine. The penetration plunger is seated at the centre of the specimen and is brought in contact with the top surface of the soil sample by applying a seated load of 4kg.

5. The dial gauge of the proving ring (for load reading) and the penetration dial gauge reading are set to zero.

6. The load is applied through the penetration plunger at a uniform rate of 1.25mm per-minute by setting the gear at constant rate of 1.25mm/min.

7. The load reading are recorded at penetration readings of 0.0, 0.5, 1.0, 1.5, 2.0, 2.5, 3.0, 3.5, 4.0, 5.0, 7.5, 10.0 and 12.5mm. In case the load readings start decreasing before 12.5mm penetration, the maximum load value and the corresponding penetration value are recorded. After the final reading, the load is released and the mould is removed from the loading machine.

8. The proving ring calibration factor is noted so that the load dial values can be converted into load in kg also find the least count of penetration dial gauge reading to convert into mm. (The present our laboratory machine, the proving ring calibration factor = 0.99 kg/div., the deflection dial gauge L.C = 0.01 mm/div.)

9. About 50 gm of soil is collected from the top three cm depth of the soil sample for the full determination of moisture content (w). Also weigh the mould with full soil and empty and determine dry density (γ_d).

Formula used in determination of $\gamma_d = \gamma_m / (1 + w/100)$ g/cc

Where, γ_m = wet density

$$\gamma_m = (W - W_m) / V_m, \text{ g/cc}$$

W = Weight of mould with moist compacted soil in gm.

W_m = Weight of empty mould in gm.

V_m = Volume of the mould in cc.

N.B. - In case of unsoaked test, (i) and (ii) are performed.

Precautions:

- i) The holes of the base plate of the mould should not be blocked.
- ii) The surcharge weight should be aligned with the plunger so that the plunger penetrates freely into the soil.

Formulae to calculate the expansion ratio and CBR value:

The swelling or expansion ratio is calculated from observation during the swelling test using this formula:

$$\text{Expansion ratio} = 100(df - dt)/h$$

Where df = Final dial gauge after soaking in mm.

dt = initial dial gauge reading before soaking in mm

h = Initial height of the specimen in mm.

The CBR value is calculated from this formula,

$$\text{CBR}\% = [\text{Load carried by soil sample at 2.5 or 5mm penetration} / \text{Load carried by standard crushed stone at 2.5 or 5mm penetration}] \times 100$$

Results:

The expansion ratio of soil due to soaking and other details of the test may be reported as given in the observation sheet. The CBR values at 2.5 and 5mm penetrations are calculated for each specimen from the corresponding graphs as shown in Fig.1.2. Generally the CBR value at 2.5mm penetration is higher and this value is adopted. However if the CBR value is obtained at 5mm penetration, the test is to be repeated to verify

the results. If the value at 5mm is again higher than it is adopted as the CBR value of the soil sample. The average CBR value of three specimens is reported to the first decimal place.

Fig.1.2: Typical load penetration curve

According to the Indian Road Congress, if the maximum variation in laboratory in CBR values between the three specimens exceed the value given below for the different ranges, the CBR test should be repeated on additional three specimens and the average value of six specimens is adopted.

Maximum permissible variation in CBR value %	Range of CBR value , %
3.0	Up to 10
5.0	10 to 30
10.0	30 to 60
Not significant	Above 60

Record of Observations:

Compaction moisture content =

Dry density =

Condition of test specimen: soaked/unsoaked

Moisture content: a) At top 3cm layer after soaking =

b) Average after soaking =

Proving ring calibration factor =

Surcharge weight =

Period of soaking =

Expansion ratio =

Sample no (1)	Dial gauge reading in div. (2)	Penetration in mm (3)	Proving ring dial gauge reading in div. (4)	Load on plunger in kg	CBR value at 2.5 and 5 mm from graph (6)
1	0 50 100 150 200 250 300 400 500 750 1000 1250	0 0.5 1.0 1.5 2.0 2.5 3.0 4.0 5.0 7.5 10.0 12.5			CBR at 2.5 mm = CBR at 5mm =

(Table for sample 2)

(Table for sample 3):

Average CBR value at penetration 2.5 mm = %

Average CBR value at penetration 5 mm = %

CBR of the sample (to be adopted for design) = %

Results: The mean CBR value of the three samples is the CBR value of the subgrade.

Experiment no – 8

Title : Dynamic Cone Penetration Test [as per ASTM D6951/D6951M-09]

Aim : Determination of Dynamic Cone Penetration Test

1. SCOPE

This method describes the determination of the rate of penetration of the Dynamic Cone Penetrometer (DCP) into a natural or compacted material by virtue of the built-in sliding hammer. The penetration rate is inversely proportional to the resistance of the ground to the penetration of the cone of the DCP and may be related, inter alia, to the in situ CBR or soil density

2. APPARATUS

1. Dynamic Cone Penetrometer as illustrated in Figure ST6/1 with the appropriate spanners, spare cones, rods, etc. (See 5.2)
2. A pick or hand auger.
3. A spade.
4. A measuring tape, 2m long.
5. Traffic cones, warning signs and flags as required

METHOD

Assemble the DCP as shown in Figure ST6/1 ensuring that the parts are fitting properly and that the hammer can slide freely. Place the tip of the cone on the site to be tested (see 5.2). Hold the DCP vertically and by means of a hammer knock the cone into the surface up to the zero mark, which is the parallel-sided shoulder portion (=3mm wide) just above the cone-shaped tip.

Attach the measuring rod to the DCP and zero the sliding scale.

While holding the DCP vertically, lift the hammer as far as it can go and allow it to fall freely and strike the anvil, driving the cone into the ground surface.

The penetration can be read off after each blow of the hammer or after as many blows as are practical or required for the purpose of the test (see 5.3). Record the penetration (to the nearest 1mm) and the number of blows on form ST6/1 or a similar forms.

On completion of the test, the DCP is extracted by ramming the hammer against the upper stop - usually after the measuring scale has been detached to prevent damage (see 5.5).

The strength of layers deeper than the reach of the DCP can be measured by removing some

or all of the overlying material with a pick and spade or using a hand auger. At the start of the test the depth below the original datum level of the material to be tested is measured, using a trace measure and recorded.

CALCULATIONS

4.1 The DCP penetration depths in mm are plotted against the number of blows on Form ST6/1 and a penetration curve is drawn, the angle of which is the penetration rate, known as the “DCP number” (DN) in mm/blow. A consistent slope angle thus indicates a consistent DN for that particular zone.

4.2 In situ shear strength (CBR)

From the curve, the DN (in mm/blow) at different zones or the mean over a chosen depth may be calculated to obtain the in situ shear strength of the particular zone. The DN of a specific zone is obtained by dividing the zone thickness by the number of hammer blows to penetrate that zone.

From Table 1 the equivalent in situ CBR value of that zone can be obtained

TABLE 1

DCP number (mm/BLOW)	(IN SITU CBR	DCP number (mm/ BLOW)	<i>IN SITU CBR</i>
3	110	17	12
4	75	18	11
5	55	19	10
6	45	20	9
7	35	22	8
8	30	25	7
9	25	28	6
10	22	32	5
11	20	38	4
12	18	48	3
13	16	60-70	2
14	15	+80	1
15	14		

NOTES

1 The DCP may be used to assess the density of a fairly uniform material by relating density to penetration rate (DN) on the same material. In this way under-compacted or “soft spots” can be identified even though the DCP does not measure density directly.

A field DCP measurement results in a field or in situ CBR and will normally not correlate with the laboratory or soaked CBR of the same material. The test is thus intended to evaluate the in situ strength of a material under prevailing conditions.

If a number of DCP penetrations have to be compared with one another (or with target values), or when the mean values for a section of road have to be estimated, the DN values may be recorded on the DCP layer-strength diagram as illustrated in Form ST6/2.

Experiment no : 9

Title : Penetration test of bitumen

Aim: To determine the Penetration value of Bitumen

Apparatus:(i) Container

(ii) Needle

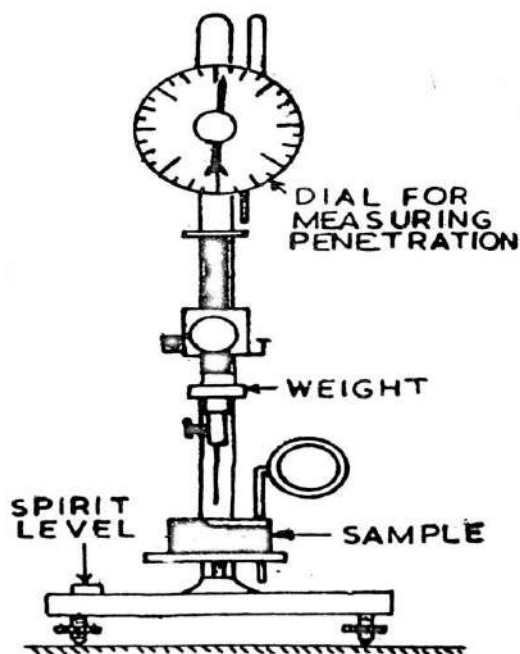
(iii) Water bath

(iv) Tray

(v) Penetration Apparatus

(vi) Thermometer

(vii) Time measuring device



Theory: The consistency of bituminous materials vary depending upon several factors such as constituent, temperature, etc. The consistency of bitumen is determined by penetration test which is a very simple test. Various types of grades of bituminous materials are available depending on their origin and relining process. The penetration test determines the consistency of these materials for purpose of grading them, by measuring the depth in tenths of mm of which a standard needle will penetrate vertically in 5 second under specified condition and temperature. Thus the basic principle of the penetration test is the measurement of the penetration (in units of one tenth of a mm) of a standard needle in a bitumen sample maintained at 25⁰ C during 5 seconds the total weight of the needle assembly being 100g. The softer the bitumen, the greater will be the penetration. The test is widely used world over for classifying the bitumen into different grades.

Procedure:

- i. Penetration of test specimen: soften the material to a pouring consistency at a temperature not more than 60⁰C for tars and 90⁰C bitumen above the approximate softening point and stir it

thoroughly until it is homogeneous and is free from air bubbles and water. Pour the melt into the container to a depth of at least 10mm in excess of the expected penetration. Protect the sample from dust and allow it to cool in an atmosphere at a temperature between 15°C to 30°C for one hour. Then place it along with the transfer dish in the water bath at $25.0^{\circ}\pm 0.1^{\circ}\text{C}$ and allow it to remain for 1 to 1 and half hour. The test is carried out at $25.0^{\circ}\pm 0.1^{\circ}\text{C}$ unless otherwise stated.

- ii. Fill the transfer dish water from the water bath to depth sufficient to cover the container completely. Place the sample in it and put it open the stand of the penetration apparatus.
- iii. Clean the needle with benzene, dry it and load with the weight. The total moving load required is 100 ± 0.25 gms, including the weight of the needle, carrier and superimposed weights.
- iv. Adjust the needle to make contact with the surface of the sample. This may be done by placing the needle point in contact with its image reflected by the surface of the bituminous material.
- v. Make the pointer of the dial to read zero or not the initial dial reading.
- vi. Release the needle for exactly five seconds.
- vii. Adjust the penetration machine to measure the distance penetrated.
- viii. Make at least three readings at point on the surface of the sample not less than 10mm apart and not less than 10mm from the side of the dish. After each test return the sample and transfer dish to the water bath and was the needle clean with benzene and dry it. In case of material of penetration greater than 225 three determination on each of the two identical test specimens using a separate needle for each determination should be made, leaving the needle in the sample on completion of each determinations to avoid disturbance of the specimen.

Experiment no : 10

Title : Ductility Test

Aim: To determine of Ductility value of Bitumen sample

Apparatus uses:

- I. Briquette mould
- II. Water bath
- III. Testing machine
- IV. Thermometer

Theory : In the flexible pavement construction where Bitumen binders are used, it is significant importance that the binders form ductile theme films around the aggregates. This serves as a satisfactory binder in improving the physical in interlocking of the aggregates. Under traffic loads the bituminous pavement layer is subjected to repeated deformation and recoveries. The binder material which does not possess sufficient ductility would crack and thus provide previous pavement surface. Ductility test is carried out on Bitumen to test this ductile property of the binder.

The test is believed to measure the adhesive property of bitumen and its ability to stretch. The ductility is expressed as the distance in cm to which a standard briquette can be stretched before the thread breaks. The test is conducted at 27°C and at the rate of pull of 50mm/min.



Procedure:

- i. Melt the bituminous test material completely at a temperature of 75°C to 100°C above the approximate softening point until it becomes thoroughly fluid.
- ii. Strain the fluid through IS sieve 30.
- iii. After stirring the fluid, pour it in the mould assembly and place it on a brass plate.
- iv. In order to prevent the material under test for sticking, coat the surface of the plate and interior surface of the slides of the mould with mercury or by a mixture of equal parts of glycerin and dextrin.
- v. After about 30-40 minutes, keep the assembly along with the sample in a water bath. Maintenance the temperature of the water bath at 27°C for half an hour.
- vi. Remove the sample and mould assembly from the water bath and trim the specimen by leveling the surface by using a hot knife.
- vii. Replace the mould assembly in water bath maintenance at 27°C for 80 to 90 min.
- viii. Removes the slides of the mould
- ix. Hook the clips carefully on the machine without causing any initial strain.

- x. Adjust the pointer to read zero.
- xi. Start the machine and pull two clips horizontally at a speed of 50mm per minute.
- xii. Note the distance at which the bitumen thread of specimen breaks.
- xiii. Record the observations in the performance and compute the ductility value. Report the mean of two observations rounded to nearest whole number as the ductility value.

Observation:

- i. Bitumen grade =
- ii. Pouring Temperature °C =
- iii. Test Temperature °C =
- iv. Periods of cooling, minutes =
 - a) In air =
 - b) In water bath before trimming =
 - c) In water bath after trimming =

Test property	Briquette No.		Mean value
	1	2	
a) Initial reading			
b) Final reading			
c) Ductility= b-a cm			

Results: The suitability of bitumen is judged depending upon its type and proposed use. Bitumen with low ductility value may get cracked especially in cold weather. ISI has specified following values of minimum ductility for various grades of bitumen as follows:

Source of paving bitumen	Penetration grade	Minimum ductility value cms
Assam petroleum	A-25	5
	A-35	10
	A-45	12
	A-65, A-90 & A-200	15
other	S-35	50
	S-45, S-65 & S-90	75

Experiment No. 11

Title : Determination of softening point of Bitumen using Ring and Ball Apparatus.

Aim: The aim of the experiment is to determine the softening point of the bituminous material.

Apparatus used:

The Ring & Ball apparatus consist of following:

- **Steel ball:** Two members of each of 95 mm diameter and weighting (3.5±0.05)g.
- **Brass ring:** Two numbers each having depth of 6.40mm. The inside diameter at bottom and top is 15.90mm and 17.50mm respectively.
- Ball guides to guides the movement of steel balls centrally.
- **Support:** That can be hold rings in position and also allows for suspension of a thermometer.
- Thermometer that can read up to 100°C with an accuracy.
- **Bath:** A heat resistant glass beaker not less than 85 mm in diameter and 1220 mm in depth.

Theory:

Bitumen does not suddenly change from solid to liquid state, but as the temperature increase it is gradually become softer unit it flows radially. Softening point is essentially the temperature at which the bituminous binders have an equal viscosity. In other words, the softening point is the temperature at which the substance attains particular degree of softening under specified condition of test. For bitumen, it is obvious that harder grade bitumen higher softening point than softer grade bitumen.

Procedure:

- 1) Preparation of test sample: Heat the material to a temperature between 75°C -100°C above its softening point. Stir until it is completely fluid and free from air bubble and water, if necessary. Filter it through IS Sieve 30. place the rings previously heated to a temperature approximating to that of the malcontent materials. On a metal plate, which has been coated with a mixture of equal parts of glycerine and doctrine After cooling for 30 minutes in air, level the material in the ring by removing the excess with a warmed sharp knife.
- 2) Assemble the apparatus with the ring's thermometer and ball guiders in position.
- 3) Fill the bath with distilled water to a height of 50 mm above the upper surface of the rings. The starting temperature should be 5°C.

Note: Use glycerine in place of water if the softening point is expected to be above 80°C

- 4) Apply heat to the bath and stir the liquid so that the temperature rises at uniform rate of 5±.05 (min)
- 5) As the temperature increase the bituminous materials softens and the ball sinks through the ring carrying a portion of the material with it.
- 6) Note down the temperature when any one of the steel balls with bituminous coating touches the bottom plate.

- 7) Record the temperature when the second ball also touches the bottom plate. The average of the two reading is recorded as the softening point.

Temperature when the ball touches bottom C	Temperature	Average Temperature
A		
B		

Experiment No - 12

Aim : Determine the Flash and Fire point of bituminous material using Pansky – Martens Aparatus

The fire point is the lowest temperature at which the materials ignited and burns and specified Condition of test.

Procedure:

- (i) Clean and dry all parts of the cup and its accessories thoroughly.
- (ii) Fill the cup with the material to be tested up to the level indicated by the billing mark.
- (iii) Place the lid on the Cup and set the latter in the stove.
- (iv) Insert the thermometer.
- (v) Light and adjust the test flame so that it is of the Size of bead of 4mm in dia. Applied heat such that the temperature rises at the rate of 5°-6° /min.
- (vi) Turn the stirrer at a rate of approximately 60 rpm/min.

Apply the test flame by operating the device controlling the shutter and test flame burner so that the flame Is lowered in 0.5 sec. Left in its lower position. Discontinue stirring during the application of test flame.

(vii) Apply the test flame Initially at a temperature 17°C below the expected flash point. There after apply the test flame at an interval of 1°C for the range above 104°C increase this interval to 20°C.

(viii) Note down the flash point as the Temperature of which the flame application causes a distinct flash point in the interior of the cup.

The duplication test results should not differ after by more than the following:-

Observation:

Type of material:- MC-4

Type of test:- Closed.

Property	Test			Mean
	1	2	3	
Flash point	190 °	180°	-	185°
Fire point	240°	215°	-	227.5°

Experiment no:13

Aim: Determination of stripping value of road aggregates.

Apparatus uses:

1. Thermostatically controlled water bath.
2. Beakers of capacity 500 ml.



Fig:Thermostatically controlled water bath.

Definition:

The stripping value of aggregates is determined as the ratio of the uncovered area observed visually to the total area of aggregates, expressed as a percentage.

Procedure:

The aggregate sample the test sample consists of aggregate of size passing 25 mm sieve and retained on 12.5 mm sieve.

- (i) Obtain the material that passes through 25 mm sieve and is retained on 12.5 mm sieve.
- (ii) Dry clean and heat the binder and aggregates and aggregates to 150°C -175 °C to120°C -150°C respectively and mix with 5% binder by weight of aggregate.
- (iii) After complete coating, allow the mixture to cool at room temperature in clean dry beaker.
- (iv) Add distilled water to immerse the coated aggregates.
- (V) Cover the beaker and keep it undisturbed in a thermostatic water bath at a temperature of 40°C for a period of 24 hrs.
- (vi) Estimate the extent of stripping by visual examination while the specimen is under water and express as the average per cent area of aggregate surface uncoated.

Observations:

	Sample 1	Sample 2	Sample 3
Percentages of area of Aggregate uncoated by immersion in water	%	%	%
Average stripping = %			

Results:

The results of stripping test give an indication regarding susceptibility of aggregates to the action of water, or moisture. The more the stripping value, the poorer are the aggregates from point of view of adhesion. IRC has specified the maximum stripping value of 25% for aggregates to be used in bituminous road construction.

Experiment no :14

Aim: To determine the viscosity of bitumen.

Theory : Viscosity is defined as inverse of fluidity. Viscosity thus defines the fluid property of bituminous material. The degree of fluidity at the application temperature greatly influences the ability of bituminous material to spread, Penetration into the voids and also coat the aggregates and hence affects the strength characteristics of the resulting paving mixes.

Apparatus: Measuring cylinder, stop watch, soap water.

Procedure:

1. Take 50 ml soap water.
2. Weight 2 gm of bitumen and melt it 70c-90c.
3. Pour 50 ml soap water in a measuring cylinder.
4. Pour the bitumen sample on the measuring cylinder and note the time taken by the bitumen sample to settle down.

Result: The time in seconds for 50ml of the test sample to flow in the measuring cylinder is define as the viscosity at a given test temperature.

EXPERIMENT NO. 15

MARSHALL STABILITY TEST FOR BITUMEN MIX DESIGN

Objective: To find out optimum bitumen content of given bituminous mix.

Theory:

Bruce Marshall, formerly bituminous engineer with Mississippi state highway department, USA formulated Marshall's method for designing bituminous mixes. Marshall's test procedure was later modified and improved upon by U.S. corps of engineers through their extensive research and correlation studies. ASTM and other agencies have standardized the test procedure. Generally, this stability test is applicable to hot-mix design using bitumen and aggregates with maximum size of 25mm.

In this method, the resistance to plastic deformations of cylindrical specimen of bituminous mixture is measured when the same is loaded at the periphery at 5 cm per minute. This test procedure is used in designing and evaluating bituminous paving mixes. The test procedure is extensively used in routine test program for the paving jobs. There are two major features of the Marshall method of designing mixes namely, (i) Density-voids analysis, (ii) stability-flow tests. The Marshall stability of the mix is defined as a maximum load carried by a compacted specimen at a standard test temperature at 60°C. The flow value is a deformation the Marshall Test specimen undergoes during the loading up to the maximum load in 0.25 mm units. In this test an attempt is made to obtain optimum binder content for the type of aggregate mix and traffic intensity. The proposed designed steps for the design of bituminous mix are given below:

- i) Select grading to be used.
- ii) Select aggregates to be employed in the mix.
- iii) Determine the proportion of each aggregate required to produce design grading.
- iv) Determine the specific gravity of the aggregate combination and of the asphalt cement.
- v) Make up trial specimens with varying asphalt contents.
- vi) Determine the specific gravity of each component specimen.
- vii) Make stability tests on the specimens.
- viii) Calculate the percentage of voids, VMA and the percent voids filled with bitumen each specimen.
- ix) Select the optimum bitumen content with design requirements. The design may be required if necessary after altering the gradation so as to fulfil the design requirements.

Apparatus:

1. *Mould assembly*: Cylindrical moulds of 10cm diameter and 7.5cm height are required. It further consist of a base plate and collar extension. They are designed to be interchangeable with either end of cylindrical mould.

2. *Sample Extractor*: For extruding the compacted specimen from the mould, an extractor suitably fitted with a jack or compression machine.

3. *Compaction pedestal and hammer*: It consist of a wooden block capped with M.S. plate to hold the mould assembly in position during compaction. The compaction hammer consist of a flat circular tamping face 8.8 cm diameter and equipped with a 4.5 kg. Weight constructed to provide a free fall of 47.5cm. Mould holder is provided consisting of spring tension device designed to hold compaction mould in place on the compaction pedestal.

4. *Breaking head*: It consist of upper and lower cylindrical segments or test heads having an inside radius of curvature of 5cm. The lower segment is mounted on a base having two vertical guide rods which facilitate insertion in the holes of upper test head.

5. *Loading machine*: The loading machine is provided with a gear system to lift the base in upward direction. On the upper end of the machine, a pre-calibrated proving ring of 5 tonne capacity is fixed. In between the base and the proving ring, the specimen contained in test head is placed. The loading machine produces a movement at the rate of 5cm per minute. Machine is capable of reversing its movement downward also. This facilitates adequate space for placing test head system after one specimen has been tested.

Loading machine set up

Flow Meter: One dial gauge fixed to the guide rods of a testing machine can serve the purpose. Least count of 0.025 mm is adequate. The flow value refers to the total vertical upward movement from the initial position at zero load to a value at maximum load.

The dial gauge or the flow meter should be able to measure accurately the total vertical movement upward. Besides the above equipment, the following are also required.,

- i) Ovens on hot plate,
- ii) Mixing apparatus,
- iii) Water bath, thermometers of range up to 200°C with sensitivity of 2.5°C. Bulk density determination,
- i) Stability and flow test, ii) Density and voids analysis, iii) Balance with an accuracy of 0.1 gm.

Procedure:

In the Marshall method each compacted test in specimen is subjected to the following tests and analysis in the order listed below:

At least three samples are prepared for each binder content.

Preparation of test specimens:

The coarse aggregates, fine aggregates and the filter material should be proportioned and mixed in such a way that final mix after blending has the gradation within the specified Range. The specified gradation of mineral aggregates and the bitumen binder as per IRC: 29-1968 are given in table 10.1.

Table 10.1: Composition of dense graded bituminous macadam Sieve size	Percentage passing by	Percentage passing by
	weight grade 1	weight grade 2
20mm	-	100
12.5mm	100	80-100
10mm	80-100	70-90
4.75mm	55-75	50-70
2.36mm	35-50	35-50
600μ	18-29	18-29
300 μ	13-23	13-23
150 μ	8-16	8-16
75 μ	4-10	4-10
Binder content percent by weight of mix	5-7.5	5-7.5